State-dependent splitting errors for anisotropic transport and diffusion: Counterexamples and a conditional corrected bound

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28 October 2026

Abstract

We audit the nine-page arXiv version 2603.08799v1 of Zylberman, Fredon, Loureiro, and Debbasch. Two explicit smooth periodic examples contradict, respectively, the assertion that all derivative norms are conserved by the allowed transport flows and the printed leading diffusion splitting coefficient. The diffusion example has strictly positive coefficients and positive, nonzero-mean initial data. We derive the missing mixed derivatives, explain the distinction between a loose operator-norm comparator and an algorithmic speedup, and document regularity, normalization, and resource assumptions that need further justification. As a replacement framework, we prove an exact autonomous finite-dimensional splitting bound with an explicit uniform commutator hypothesis and a normalized-state corollary. Deterministic classical checks accompany the note. We neither claim to have found every error nor prove that mesh-independent vector-norm scaling is impossible. This is a correction and reproducibility study, not a new end-to-end quantum algorithm or a claim of priority.

1 Scope, provenance, and standards of evidence

The audited object is the local nine-page PDF identifying itself as arXiv:2603.08799v1, submitted 9 March 2026 [1]. Its SHA-256 is `d03a710413b32ac622700adcfbaa9c3b860 4ab9db00e45c348ea06c3d4be8988`. The source record links a published conference chapter [3]; we checked its public metadata, but did not compare the full chapter text. No finding here is asserted to apply unchanged to an unexamined revision, a later journal treatment, or other work by the same authors.

We distinguish four evidence categories:

1. **Demonstrated error:** an identity, inequality, or prescription fails as written, with an explicit calculation.
2. **Missing hypothesis or proof gap:** the intended result may hold under additional assumptions, but the displayed argument does not establish it.
3. **Discretization limitation:** a property of the selected scheme that does not necessarily destroy its intended convergence regime.
4. **Editorial ambiguity:** notation or indexing that should be repaired without implying a substantive mathematical failure.

The motivation for vector-norm analysis remains sound. An, Fang, and Lin previously studied circumstances in which state-dependent Trotter estimates avoid large discretized operator norms [2]. A failed prefactor does not by itself invalidate that research program.

2 Model and consistent notation

On the unit periodic torus $\mathbb{T}^d$, consider separately

$$\partial_t f + \sum_{j=1}^{d} c_j(x,t)\partial_j f = 0, \tag{1}$$

$$\partial_t u = \sum_{j=1}^{d} \kappa_j(x,t)\partial_j^2 u. \tag{2}$$

The source imposes $\partial_j c_j = \partial_j \kappa_j = 0$. Under this restriction, (2) equals the divergence-form equation in its Eq. (1). For forward diffusion we explicitly require $\kappa_j \geq 0$. The stronger ellipticity condition $\kappa_j \geq \kappa_* > 0$ is useful for some regularity arguments, but is not needed for the elementary finite-grid contraction below.

Let $M_j = 2^{n_j}$, $h_j = M_j^{-1}$, and $M = \prod_j M_j$. The total number of data qubits is $n_{\text{tot}} = \sum_j n_j$. A grid coordinate is $x_j = r_j/M_j$, $0 \leq r_j < M_j$; r_j is the integer register label. We use the quadrature norm

$$\|v\|_h^2 = M^{-1} \sum_{\boldsymbol{r}} |v_{\boldsymbol{r}}|^2.$$

This converges to the continuum L^2 norm for smooth data on the unit torus. The associated unit quantum state has amplitudes $v_{\boldsymbol{r}}/(\sqrt{M}\,\|v\|_h)$. An unnormalized grid solution and its normalized quantum encoding are different objects and will not share a symbol silently.

For centered first differences write

$$\delta_{p,h} v = h^{-1} \sum_{k=-p}^{p} a_k S^k v, \qquad a_{-k} = -a_k, \qquad D_{p,h} = -i\delta_{p,h}.$$

Periodic shifts make $D_{p,h}$ Hermitian. Because the coefficient in direction j is independent of that direction, its multiplication matrix commutes with D_j. Consequently $-ic_j D_j$ is skew-Hermitian and $-\kappa_j D_j^2$ is Hermitian negative semidefinite when $\kappa_j \geq 0$. The source's directional exponentials then yield unitary transport factors and contractive, generally non-unitary diffusion factors.

3 A transport counterexample and energy estimate

3.1 Derivative norms are not generally conserved

Page 6, footnote 10, asserts conservation of the L^2 norm of any spatial derivative of f. The coefficient conditions stated on page 2 do not imply this. Choose the allowed time-independent periodic shear

$$c(x,y) = (\sin(2\pi y), 0), \qquad f_0(x,y) = \sin(2\pi x).$$

Both components are smooth and independent of their own coordinate. The exact solution is

$$f(x,y,t) = \sin\big(2\pi[x - t\sin(2\pi y)]\big).$$

Direct differentiation and integration give

$$\|f(t)\|_{L^2} = 1/\sqrt{2}, \tag{3}$$

$$\partial_y f = -4\pi^2 t \cos(2\pi y)\cos\big(2\pi[x - t\sin(2\pi y)]\big), \tag{4}$$

$$\|\partial_y f(t)\|_{L^2} = 2\pi^2 |t|. \tag{5}$$

The derivative norm starts at zero and grows. This is an analytic counterexample, not a conclusion inferred from a numerical plot.

3.2 What can replace conservation

The velocity is divergence-free because $\sum_j \partial_j c_j = 0$. For a smooth solution, differentiation of (1) gives

$$(\partial_t + c \cdot \nabla)\nabla f = -(\nabla c)^{\mathsf{T}} \nabla f,$$

where $(\nabla c)_{ij} = \partial_j c_i$. Periodic integration by parts yields

$$\frac{\mathrm{d}}{\mathrm{d}t} \|\nabla f(t)\|_{L^2}^2 \le 2 \|\nabla c(t)\|_{L^\infty,\mathrm{op}} \|\nabla f(t)\|_{L^2}^2 .$$

Thus

$$\|\nabla f(t)\|_{L^2} \le \exp\!\left(\int_0^t \|\nabla c(s)\|_{L^\infty,\mathrm{op}} \,\mathrm{d}s \right) \|\nabla f_0\|_{L^2} .$$

This bound involves the initial full gradient and the velocity gradients; it does not conserve each coordinate derivative separately. Higher Sobolev estimates require corresponding coefficient derivatives. Constants can therefore depend on the flow and on T, even when they are independent of h.

3.3 Directional commutators

For $L_j = c_j \partial_j$ and $L_m = c_m \partial_m$,

$$[L_j, L_m] f = c_j (\partial_j c_m) \partial_m f - c_m (\partial_m c_j) \partial_j f.$$

The transport prefactor on page 6 has this appropriate first-derivative structure. Its use of a supremum over evolved derivatives can remain meaningful. The invalid step is identifying those derivative norms with their initial coordinatewise values.

4 Diffusion: missing terms and a counterexample

4.1 Full continuum commutator

Put $A_j = \kappa_j \partial_j^2$. Expanding products of differential operators for $j \neq m$ gives

$$\begin{aligned} [A_j, A_m] u = {} & \kappa_j (\partial_j^2 \kappa_m) \partial_m^2 u + 2\kappa_j (\partial_j \kappa_m) \partial_j \partial_m^2 u \\ & - \kappa_m (\partial_m^2 \kappa_j) \partial_j^2 u - 2\kappa_m (\partial_m \kappa_j) \partial_m \partial_j^2 u. \end{aligned} \tag{6}$$

The common fourth-derivative term cancels. The mixed third derivatives do not cancel under $\partial_j \kappa_j = 0$. The displayed a'_g following the source's Eq. (8), page 6, contains only the first and third terms of (6) in its norm estimate.

A valid direct continuum estimate is

$$\begin{aligned} \|[A_j, A_m] u\|_{L^2} \le {} & \|\kappa_j\|_\infty \left\| \partial_j^2 \kappa_m \right\|_\infty \left\| \partial_m^2 u \right\|_{L^2} \\ & + 2 \|\kappa_j\|_\infty \|\partial_j \kappa_m\|_\infty \left\| \partial_j \partial_m^2 u \right\|_{L^2} \\ & + \|\kappa_m\|_\infty \left\| \partial_m^2 \kappa_j \right\|_\infty \left\| \partial_j^2 u \right\|_{L^2} \\ & + 2 \|\kappa_m\|_\infty \|\partial_m \kappa_j\|_\infty \left\| \partial_m \partial_j^2 u \right\|_{L^2} . \end{aligned} \tag{7}$$

It requires control of additional derivatives. This is an identity and triangle-inequality bound in the continuum, not by itself a theorem about all discrete split trajectories.

4.2 Strictly positive coefficients and positive initial data

On $\mathbb{T}^2$ choose, for $0 < a, b < 1$ and integer $m \geq 2$,

$$\kappa_x = 1, \qquad \kappa_y = 1 + a\sin(2\pi x), \qquad u_0 = 1 + b\cos(2\pi m x)\cos(2\pi y).$$

The coefficients satisfy all source coordinate restrictions and are strictly positive. The initial data are positive with mean 1. Let $q = 2\pi$, $A = \partial_x^2$, $B = \kappa_y \partial_y^2$. Then, with the commutator convention $[A, B] = AB - BA$,

$$[A, B]u_0 = abq^4 \left[(m + \tfrac{1}{2})\sin(q(m+1)x) + (m - \tfrac{1}{2})\sin(q(m-1)x)\right]\cos(qy).$$

Fourier orthogonality gives

$$\|u_0\|_{L^2}^2 = 1 + b^2/4, \qquad \|[A, B]u_0\|_{L^2} = \frac{abq^4}{2\sqrt{2}}\sqrt{4m^2 + 1}.$$

The quantity built from the two retained terms in the source, before the common factor $1/2$ and normalization, is only

$$\|\kappa_x\|_\infty \left\|\partial_x^2 \kappa_y\right\|_\infty \left\|\partial_y^2 u_0\right\|_{L^2} = abq^4/2.$$

The ratio of the actual commutator norm to that printed comparator is

$$R_m = \sqrt{(4m^2 + 1)/2} > 1.$$

For $m = 4$, $R_m = \sqrt{65/2} \simeq 5.700877$.

4.3 Why normalization does not remove this counterexample

For real self-adjoint A, B, the commutator is skew-adjoint, so $\langle u_0, [A, B]u_0\rangle = 0$. The differential of $N(v) = v/\|v\|$ at u_0 is

$$DN_{u_0}(w) = \frac{(I - \hat{u}_0\hat{u}_0^{\mathsf{T}})w}{\|u_0\|}, \qquad \hat{u}_0 = u_0/\|u_0\|.$$

The leading splitting defect lies perpendicular to u_0, and survives normalization unchanged apart from division by $\|u_0\|$.

To connect this continuum example to the source's actual stencil, set $A_\eta = -D_x^2$ and $B_\eta = -K_y D_y^2$ on sufficiently fine centered periodic grids. Acting on this fixed trigonometric data and coefficient family, the discrete derivatives and their commutators converge in quadrature norm to the displayed continuum expressions. The printed initial-time comparator converges to the smaller expression above. Both matrices are real self-adjoint, so the discrete orthogonality also holds exactly. Thus the strict ratio persists on sufficiently fine fixed grids, before taking the time-step limit.

On each such fixed finite grid, the global first-order Lie-splitting expansion is an ordinary matrix Taylor calculation:

$$\lim_{L\to\infty} L(S_{T/L}^L - U(T))u_0 = \frac{T}{2}\int_0^T U(T - s)[B, A]U(s)u_0\,\mathrm{d}s,$$

where $U(t) = e^{t(A+B)}$ and $S_h = e^{hB}e^{hA}$. Here A, B denote the fixed-grid matrices just defined. Their autonomous exponentials are analytic, so the matrix expansion is justified. As $T \downarrow 0$, the normalized leading coefficient divided by T^2 approaches $\|[B, A]u_0\|/(2\|u_0\|)$. The printed derivative-only prefactor approaches the smaller comparator above divided by $2\|u_0\|$. Thus the claimed leading coefficient fails for sufficiently small positive T, if the remainder is genuinely $o(1/L)$ as the leading-error interpretation requires. If a remainder is allowed to contain another $1/L$ contribution, the advertised prefactor has not been isolated. This argument concerns a coefficient, not an impossibility of mesh-independent bounds with (7) and adequate regularity.

5 Other errors, limitations, and unresolved proof obligations

5.1 Positivity is an essential explicit hypothesis

Page 5, footnote 8, uses contraction for diffusion. This follows from $\kappa_j \geq 0$, but that inequality is not among the explicit mathematical assumptions on page 2. Negative constant $\kappa = -1$ satisfies the written smoothness and coordinate restrictions and gives backward heat: $u = e^{(2\pi)^2 t}\cos(2\pi x)$. This is a missing hypothesis; it is not a criticism of physically nonnegative thermal conductivity.

5.2 Squared centered derivatives have extra zero modes

For $p = 1$,

$$-\delta_{1,h}^2 u_r = -\frac{u_{r+2} - 2u_r + u_{r-2}}{4h^2}, \qquad \lambda(\theta) = \sin^2\theta/h^2.$$

Besides the constant mode, the checkerboard $u_r = (-1)^r$ lies in the kernel on an even grid. For every antisymmetric centered first-derivative stencil, $\sum_k a_k e^{ik\pi} = 0$. In d dimensions the tensor parity modes supply at least 2^d null modes, including with the source's admissible variable coefficients. This undermines uniform damping and some interpretations of long-time equilibrium. It does not contradict convergence on a fixed sufficiently smooth resolved input. A standard nearest-neighbor negative second difference has symbol $4\sin^2(\theta/2)/h^2$ and removes this extra one-dimensional zero. Its positivity and quantum implementation would need a separate analysis; substitution is not a free algorithmic repair.

5.3 Spatial residuals need stronger derivative control

The source's page-5 diffusion discussion describes a constant in terms of derivatives through order $2p + 1$. But squaring the first derivative gives, for $p = 1$,

$$\delta_{1,h}^2 u = u'' + \frac{h^2}{3}u^{(4)} + O(h^4).$$

The direct $O(h^{2p})$ pointwise consistency argument for the squared stencil needs order $2p + 2$ control. Parabolic smoothing or an integrated estimate might weaken initial regularity requirements, but such an argument is not supplied. We classify this as a regularity/proof gap, rather than exclude every possible low-regularity convergence theorem.

5.4 The directional error budget must sum to the target

The page-5 bound has the form $TK\sum_j \|c_j\|_\infty h_j^{2p}$. The printed prescription for each n_j assigns an error budget ϵ to every term. It ensures at most $d\epsilon$ from that inequality, not ϵ. Choose $\epsilon_j > 0$ with $\sum_j \epsilon_j \leq \epsilon$, and use

$$n_j \geq \max\left\{0, \left\lceil \frac{1}{2p}\log_2 \frac{TK\|c_j\|_\infty}{\epsilon_j} \right\rceil\right\}$$

for nonzero coefficients, with zero-coefficient directions handled separately and any practical minimum grid imposed additionally. An equal split uses $\epsilon_j = \epsilon/d$.

5.5 Normalization and sampling cannot be conflated

Continuum transport conserves its integral L^2 norm. Its values sampled on a fixed finite grid need not have exactly constant quadrature norm as the continuum profile evolves. A normal-

ized sampled state obeys an extra derivative-of-normalizer term. Alternatively one can derive a residual equation for fixed-normalizer samples and normalize only after comparison. For diffusion, writing the unnormalized ODE solution as a normalized qubit state introduces a similar distinction. Factors involving the final norm must be retained in a normalized error estimate.

A useful algebraic repair, valid in any Hilbert norm, is

Lemma 1 (Normalization stability)**.** *For nonzero v, w,*

$$\left\| \frac{v}{\|v\|} - \frac{w}{\|w\|} \right\| \leq \frac{2\,\|v - w\|}{\|w\|}.$$

Proof. Insert $v/\|w\|$, use the triangle inequality, and apply $|\,\|v\| - \|w\|\,| \leq \|v - w\|$. □

For mean-preserving diffusion on the unit torus, $\|u(T)\|_{L^2} \geq |\int_{\mathbb{T}^d} u_0|$. For a quantum-normalized input the corresponding bound must divide this mean by $\|u_0\|_{L^2}$. The source does impose a nonzero-mean condition on page 7; its later appearance is a scope restriction, not an undiscovered omission. No positive uniform lower bound follows from this route for mean-zero input.

5.6 Remainders and time dependence

A fixed-grid expansion with an $O_h(\Delta t^3)$ remainder is insufficient to establish a bound uniform over grids. Its constant may grow as $h \to 0$. Mixed third derivatives in (6), higher nested commutators, stability of partial split trajectories, and time derivatives of coefficients all need suitable uniform control. An $O(T^2/L)$ expression also does not isolate polynomial dependence on T if its prefactor itself contains evolved Sobolev norms growing with T.

5.7 What the exponential comparator does and does not show

Writing $M_{\text{axis}} = 2^n$ on an equal grid, the source's crude product bound uses $O(M_{\text{axis}}^2)$ for transport and $O(M_{\text{axis}}^4)$ for diffusion commutators. It is not a lower bound on the work actually needed by an algorithm. For fixed smooth coefficients and a fixed stencil, cancellation already improves a direct operator estimate. Indeed

$$[C_j D_j, C_m D_m] = C_j [D_j, C_m] D_m - C_m [D_m, C_j] D_j,$$

and a Lipschitz coefficient gives $\|[D_j, C_m]\| = O(1)$. Hence this operator-norm commutator bound is $O(M_{\text{axis}})$. For diffusion, $\left\|[D_j^2, K_m]\right\| = O(M_{\text{axis}})$ and

$$[K_j D_j^2, K_m D_m^2] = K_j [D_j^2, K_m] D_m^2 - K_m [D_m^2, K_j] D_j^2.$$

The corresponding bound is $O(M_{\text{axis}}^3)$. These are upper bounds, not asserted sharp rates in every case. Constant coefficients commute, giving no directional splitting error in the autonomous case. Therefore the advertised $\Theta(4^n)$ and $\Theta(16^n)$ improvements must be read as comparisons with a particular loose bound, not universal tight speedup factors over all operator-norm analyses.

5.8 Quantum resource and readout assumptions

A non-unitary contraction is not a deterministic ordinary unitary gate on a closed data register. An implementation must specify a dilation, block encoding, measurement/postselection, amplitude amplification, or another realization with its associated costs. For an ideal contraction block with normalization factor 1, success probability on a normalized input is $\|Vu\|^2$. Other normalizations or implementations change this cost.

Merely being differentiable does not uniformly bound the size of a Fourier or Walsh approximation, its coefficient-access cost, or the cost of loading arbitrary initial data. The finite-difference symbol has amplitude growing with grid refinement. At fixed stencil order its oscillation count as a function of normalized frequency does not grow, but the associated unitary exponential can become increasingly oscillatory at a fixed time. A gate-level complexity theorem must quantify those dependencies rather than appeal to differentiability alone.

For a fixed observable, an expectation $\langle \psi, O\psi \rangle$ is invariant under a global phase. In particular f and $-f$ encode the same quantum ray, whereas their signed point values differ. A position projector returns a normalized squared magnitude, not the signed physical field value. Recovering amplitudes requires an appropriate reference, controlled state preparation or overlap protocol, a phase convention, normalization information, and measurement-cost accounting. The measurement discussion on page 5 is therefore incomplete when interpreted as direct field-value recovery by a single expectation.

6 An exact conditional replacement theorem

We deliberately state a finite-dimensional autonomous result. It does not silently extend to the source's time-dependent PDEs.

For each grid η, let A_η, B_η be matrices such that e^{tA_η}, e^{tB_η}, and $U_\eta(t) = e^{t(A_\eta + B_\eta)}$ are contractions for $t \geq 0$. This includes skew-Hermitian transport generators and Hermitian negative semidefinite diffusion generators. Set $S_\eta(s) = e^{sB_\eta} e^{sA_\eta}$.

Assumption 1 (Uniform state-dependent commutator control)**.** *For some $h_0 > 0$ and $K < \infty$, normalized initial vectors $u_{\eta,0}$ satisfy, on every grid,*

$$\sup_{\substack{0 \leq t \leq T \\ 0 \leq r \leq s \leq h_0}} \left\| [B_\eta, A_\eta] e^{rB_\eta} e^{sA_\eta} U_\eta(t) u_{\eta,0} \right\| \leq K.$$

This explicitly includes intermediate split states. Smoothness of the continuum solution alone is not a proof of this uniform discrete hypothesis.

Proposition 1 (Uniform autonomous vector-norm splitting bound)**.** *Under Assumption 1, for integer L with $h = T/L \leq h_0$,*

$$\left\| S_\eta(h)^L u_{\eta,0} - U_\eta(T) u_{\eta,0} \right\| \leq \frac{KT^2}{2L}.$$

If also $\|U_\eta(T) u_{\eta,0}\| \geq m_* > 0$ uniformly in η, and the split output is nonzero, its normalized error is at most $KT^2/(m_* L)$.

Proof. Suppress the grid subscript. Direct differentiation gives

$$S'(s) - (A+B)S(s) = [e^{sB}, A] e^{sA} = \int_0^s e^{(s-r)B} [B, A] e^{rB} e^{sA} \, \mathrm{d}r.$$

Variation of constants provides the exact, remainder-free identity

$$(S(h) - U(h))v = \int_0^h \int_0^s U(h-s) e^{(s-r)B} [B, A] e^{rB} e^{sA} v \, \mathrm{d}r \, \mathrm{d}s.$$

Use $v = U(jh)u_0$ and Assumption 1. Contractivity of the left factors bounds each defect by $Kh^2/2$. Finally the algebraic telescoping identity

$$S(h)^L - U(h)^L = \sum_{j=0}^{L-1} S(h)^{L-1-j} (S(h) - U(h)) U(h)^j$$

and contractivity bound the sum by $LKh^2/2 = KT^2/(2L)$. The normalized conclusion follows from Lemma 1. □

The assumption makes the conclusion conditional but precise: there is no unspecified mesh-dependent Taylor remainder. For diffusion, (7) indicates the necessary higher-derivative quantities in a continuum analogue. Establishing uniform bounds on the actual discrete partial trajectories remains a separate task. For $d > 2$, pairwise defects can be summed only after the corresponding partial-product hypotheses are formulated. For time-dependent generators, time-ordered propagators and coefficient-time regularity must also be handled; neither extension is claimed here.

7 Reproducible checks and interpretation

The companion repository contains deterministic NumPy checks with analytic references, invariant checks, and grid/step refinement. They are classical computations; none executes a quantum circuit. The transport example tests both conservation of $\|f\|$ and nonconservation of $\|\partial_y f\|$. The diffusion example checks the full commutator, the missing-term ratio, and normalized split/exact behavior. Stencil checks verify the extra zero mode and second-order consistency on a fixed smooth Fourier mode. A negative-coefficient test illustrates the missing positivity hypothesis.

T	L	Normalized error	Error L/T^2	Comparator	Ratio
0.0002	64	1.1426e-07	182.819	38.062	4.803
0.0002	128	5.7130e-08	182.815	38.062	4.803
0.0002	256	2.8565e-08	182.813	38.062	4.803
0.0002	512	1.4282e-08	182.812	38.062	4.803
0.0005	64	5.8357e-07	149.394	38.351	3.895
0.0005	128	2.9177e-07	149.388	38.351	3.895
0.0005	256	1.4588e-07	149.385	38.351	3.895
0.0005	512	7.2941e-08	149.383	38.351	3.895

Table 1: Classical normalized splitting check: 128×128 grid, $p = 1$, $a = 0.2$, $b = 0.5$, $m = 4$, using the sine variant $u_0 = 1+b\sin(2\pi mx)\sin(2\pi y)$. This has the same analytic norm ratio as the cosine example. The comparator reconstructs the printed prefactor with the initial continuum derivative norm and the semidiscrete final norm. Finite numerical evidence supplements, rather than proves, the analytic coefficient objection.

The audit ledger records each finding's location, classification, support, and proposed repair. Editorial issues include mixing integer labels with dyadic coordinates on page 2, using n for both total and per-direction qubits, the time-order convention implicit in Eq. (6), and a primed/unprimed diffusion-prefactor inconsistency on page 6. These issues should not be promoted to independent physical or algorithmic refutations.

8 Limitations and open work

This note has not undergone external expert peer review. Its strongest conclusions are the displayed identities, analytic counterexamples, and the conditional matrix theorem with a complete proof. The numerical checks supplement those arguments; they do not prove a quantum advantage or an unconditional PDE theorem. An exhaustive error claim would be unjustified.

A full replacement algorithm still needs a mesh-uniform consistency and stability theorem for the chosen stencil, careful treatment of normalized states and zero-mean data, a time-dependent splitting analysis, explicit coefficient and state access models, non-unitary success-cost bounds, observable definitions, and an end-to-end comparison with appropriate classical algorithms. The counterexamples identify repairs needed in the audited version without asserting that these tasks cannot be completed.

References

[1] J. Zylberman, T. Fredon, N. F. Loureiro, and F. Debbasch, *Quantum algorithm for anisotropic diffusion and convection equations with vector norm scaling*, arXiv:2603.08799v1 (2026). https://arxiv.org/abs/2603.08799v1.

[2] D. An, D. Fang, and L. Lin, *Time-dependent unbounded Hamiltonian simulation with vector norm scaling*, Quantum **5**, 459 (2021). https://doi.org/10.22331/q-2021-05-26-459.

[3] J. Zylberman, T. Fredon, N. F. Loureiro, and F. Debbasch, *Quantum Algorithm for Anisotropic Diffusion and Convection Equations with Vector Norm Scaling*, conference chapter, pp. 255–263. Public source-linked version of record; full text not audited here. https://doi.org/10.1007/978-3-032-13855-2_23.